WORKED SOLUTION

STEP I 2006, Question 7

Pure • Integrals with Modulus Signs

STEP I 2006, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q7

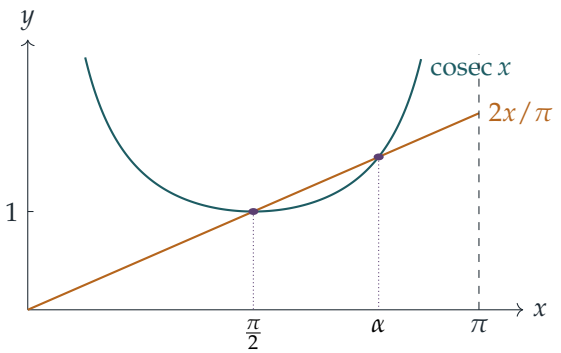
The key idea

Dividing $x \sin x = \pi/2$ by $\frac{\pi}{2} \sin x$ (which is positive on $(0, \pi)$) turns it into $\operatorname{cosec} x = 2x/\pi$, so the roots are where a straight line meets a U-shaped curve, and a line can cross a convex curve at most twice. For the integrals, the whole game with a modulus is to find where the expression inside changes sign, split the range there, and integrate each piece with the correct sign.

Solution

Part (i): the sketch and the roots

On $0 < x < \pi$, $\operatorname{cosec} x$ tends to $+\infty$ at both ends, is symmetric about $x = \pi/2$ and has its minimum value 1 there. The line $y = 2x/\pi$ runs from $(0,0)$ to $(\pi, 2)$ and passes through $(\pi/2, 1)$, the bottom of the cosec curve.



Since $\sin x > 0$ on $(0, \pi)$,

$$x \sin x = \frac{\pi}{2} \iff \frac{1}{\sin x} = \frac{2x}{\pi},$$

so the roots are the x-coordinates of the intersections in the sketch. One is visible immediately: $x = \pi/2$, since $\frac{\pi}{2}\sin\frac{\pi}{2} = \frac{\pi}{2}$.

To be sure there are exactly two, let $h(x) = \operatorname{cosec} x - 2x/\pi$. Then

$$h''(x) = \operatorname{cosec} x\,(\operatorname{cosec}^2 x + \cot^2 x) > 0 \quad \text{on } (0,\pi),$$

so h is strictly convex, and a strictly convex function has at most two zeros. (If it had three, $a < b < c$, then b would lie strictly between a and c while $h(b)$ lay on, rather than strictly below, the chord joining $(a,0)$ and $(c,0)$.) Now $h'(\pi/2) = -\cot\frac{\pi}{2}\operatorname{cosec}\frac{\pi}{2} - \frac{2}{\pi} = -\frac{2}{\pi} < 0$, so h is negative just to the right of $\pi/2$, while $h(x) \to +\infty$ as $x \to \pi^-$. By continuity h has a zero α with $\pi/2 < \alpha < \pi$. So there are exactly two roots, $\pi/2$ and α, and α is the larger.

Part (i): the integral

From the sketch (and the convexity argument), the cosec curve lies below the line between the two roots and above it beyond α. Multiplying $\operatorname{cosec} x \leqslant 2x/\pi$ by the positive quantity $\frac{\pi}{2}\sin x$, this says

$$x\sin x - \frac{\pi}{2} \geq 0 \text{ on } \left[\tfrac{\pi}{2},\alpha\right], \qquad x\sin x - \frac{\pi}{2} \leq 0 \text{ on } [\alpha,\pi].$$

Ask yourself How do we deal with the modulus sign?

We split the integral at α, where the sign changes, and on the second piece we integrate the negative of the expression.

Integrating by parts with $u = x$, $\mathrm{d}v = \sin x\,\mathrm{d}x$,

$$\int x\sin x\,\mathrm{d}x = -x\cos x + \int \cos x\,\mathrm{d}x = \sin x - x\cos x.$$

So an antiderivative of $x\sin x - \pi/2$ is

$$F(x) = \sin x - x\cos x - \frac{\pi x}{2},$$

and the integral we want is

$$I = [F(x)]_{\pi/2}^{\alpha} - [F(x)]_{\alpha}^{\pi} = 2F(\alpha) - F\left(\tfrac{\pi}{2}\right) - F(\pi).$$

The three values are

$$F(\alpha) = \sin\alpha - \alpha\cos\alpha - \frac{\alpha\pi}{2},$$

$$F\left(\tfrac{\pi}{2}\right) = 1 - 0 - \frac{\pi^2}{4},$$

$$F(\pi) = 0 - \pi(-1) - \frac{\pi^2}{2} = \pi - \frac{\pi^2}{2}.$$

Hence

$$I = 2\sin\alpha - 2\alpha\cos\alpha - \alpha\pi - 1 + \frac{\pi^2}{4} - \pi + \frac{\pi^2}{2}$$
$$= 2\sin\alpha + \frac{3\pi^2}{4} - \alpha\pi - \pi - 2\alpha\cos\alpha - 1,$$

which is the required result.

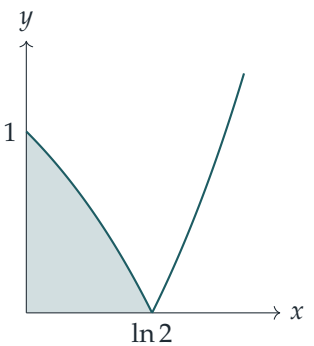
Part (ii)

Ask yourself Which of the two modulus signs actually does anything here?

On the positive x-axis, $x \geq 0$, so $e^x \geq 1$ and the inner modulus is harmless: $|e^x - 1| = e^x - 1$. The curve is therefore

$$y = |e^x - 2| \qquad (x \geq 0).$$

This equals $2 - e^x$ for $0 \leq x \leq \ln 2$, falling from $y = 1$ at $x = 0$ to $y = 0$ at $x = \ln 2$, and equals $e^x - 2$ for $x \geq \ln 2$, rising without bound.



So the curve touches the x-axis only at $x = \ln 2$, and the only region enclosed by the positive x-axis, the y-axis and the curve is the shaded one over $0 \leq x \leq \ln 2$ (beyond $\ln 2$ the curve climbs away and nothing closes up). Its area is

$$\int_0^{\ln 2} (2 - e^x)\,dx = \Big[2x - e^x\Big]_0^{\ln 2} = (2\ln 2 - 2) - (0 - 1)$$
$$= 2\ln 2 - 1 = \ln 4 - 1,$$

as required.

Checks

Bisection gives $\alpha \approx 2.4433$, and then the formula in (i) evaluates to about 0.61343; Simpson's rule applied directly to $|x\sin x - \pi/2|$ on $[\pi/2, \pi]$ gives the same value to

14 decimal places. A grid search on $(0, \pi)$ finds exactly two sign changes of $x \sin x - \pi/2$, near 1.5708 and 2.4433. In (ii), $\ln 4 - 1 \approx 0.38629$, again matching a numerical integration. As a sanity check on the sign in (i), the answer must be positive, and it is.

Common slips

- Integrating $x \sin x - \pi/2$ straight across $[\pi/2, \pi]$ without splitting at α: this gives a signed area (here negative), not the integral of the modulus.
- Missing the root at $x = \pi/2$ because the line passes through the minimum of the cosec curve, and the picture can look like a tangency. It isn't one: the curve has gradient 0 there and the line has gradient $2/\pi$, so they cross.
- In (ii), keeping both modulus signs and splitting into too many cases. Settle the sign of $e^x - 1$ first, using $x \geq 0$.

Beyond STEP

The root α has no closed form, and yet we could still compute the integral exactly in terms of it: a common situation in analysis, where a quantity is pinned down by an equation rather than a formula. The fact that a strictly convex function meets a line at most twice is one of many neat consequences of convexity, which also underlies Jensen's inequality and much of optimisation. Bartle and Sherbert's *Introduction to Real Analysis* treats continuity, the intermediate value theorem and convex functions carefully.

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